

OC Core Reviewer Attack and Response Map v1.3.3

Ontology of Continua Core release review artifact

Artifact role: Adversarial objections, boundaries, and response map

Alexander Yashin

Independent Researcher

ORCID 0009-0008-6166-0914

Version 1.3.3

Release identity: oc_core_1_3_3

Concept DOI: 10.5281/zenodo.17899134

Logion is the research-instrument and institute-automation system used to prepare, check, package, and audit the work; it is not an author.

ESTRA is the methodological framework used in the work; it is not an author or affiliation.

Dedication

Dedicated to my dear wife Maria, without whom this work would have been impossible.

Acknowledgements

Substantive review and idea acknowledgements. They are acknowledged for review pressure, ideas, criticism, or external response that improved the work. Acknowledgement does not imply authorship, endorsement, publication approval, or agreement with the theory's release form.

- G. V. Apostolov
- Eduard Fadeev
- Gennady Alekseevich Nosov
- Sergey Shpadyrev
- Stanislav Tsukrov

Abstract

This map presents the release under hostile review. It groups objections by attacked claim, explains why each attack matters, names the response route, and records residual risk and reopening conditions.

Reader Contract

Read this document by choosing the claim under attack, then following objection, response, evidence, residual risk, and reopening condition. The release promotes evidence-bound model-core claims and excludes unsupported complete-science closure, unrestricted all-domain numerical completion, or unrestricted superiority-over-modern-science claims.

Table of Contents

Dedication	1
Acknowledgements	1
Abstract	1
Reader Contract	1
1 Reader Orientation and Claim Boundaries	12
1.1 Reader Contract	12
1.2 Intended Audiences and Reading Paths	12
1.3 What OC Core Claims	13
1.4 What OC Core Does Not Claim	14
1.5 Release Scope vs Full Science Program	14
1.6 Claim Promotion, Demotion, and Background Obligations	15
2 Falsifiers and Negative Controls	16
2.1 Falsifier Registry	16
2.2 Counterexample Search	16
2.3 Known Failure Modes	17
2.4 Unsupported Strong Claims	18
2.5 Demotion Rules	18
2.6 Residual Scientific Risks	19
2.7 Full-Science Background Obligations	20
3 Novelty and Prior-Art Boundary	20
3.1 Novelty Criteria	20
3.2 Non-Equivalence Criteria	21
3.3 Overlap With Prior Art	22
3.4 Residual-Delta Analysis	22
3.5 Competitor-by-Competitor Comparison	23
3.6 Response to “This Is Just Another Theory”	24
3.7 Positioning of OC Core	24
4 Reviewer Protocol and Attack Surface	25
4.1 Review Protocol	25
4.2 Cerberus Role Map	26
4.3 Critical and High Findings	26
4.4 Closure Evidence	27
4.5 Claim Corrections	28
4.6 Hostile Reviewer Objections	28
4.7 Response Matrix	29
4.8 Remaining Non-Blocking Caveats	30
5 Versioning, Citation, and Release Identity	30
5.1 Versioning and Release Identity	30
5.2 Public Release Asset Set	31
5.3 Journal Package Map	32
5.4 Citation and Metadata Governance	32
5.5 Publication Boundary	33
5.6 Owner Approval Boundary	34
5.7 Incident and Known-Error Lessons	34
5.8 External Use Guidance	35
6 Document Boundary	36

6.1	Adversarial Review Structure and Audience	36
6.2	Limits of recovery	36
6.3	K-level criteria and falsifiers after external review	36
6.4	Falsifiability of K_{0K_0}	36
6.5	Fundamental Falsifiability Principles for K_{0K_0}	36
6.6	Structural Falsifiability Conditions	37
6.7	Boundary and Region Falsifiability	37
6.8	Energy Functional and Threshold Falsifiability	37
6.9	Cycle and Evolution Falsifiability	37
6.10	Falsifiability via Modal Spaces and Modal Dimensions	37
6.11	Falsifiability via Meta-Cycles C_{10}	38
6.12	Falsifiability via Cycles C_{11}	38
6.13	Falsifiability via Universal Cycles C_{12}	38
6.14	Percolation and Threshold Falsifiability	38
6.15	Boundary and Admissibility Falsifiability	38
6.16	Falsifiability via Chemical Potentials and Thresholds	38
6.17	Falsifiability via Cycles C_{4C_4}	39
6.18	Falsifiability via Membrane and Electrical Boundary	39
6.19	Falsifiability via Neural Cycles C_{5C_5}	39
6.20	Falsifiability via Cognitive Cycles C_{6C_6}	39
6.21	Falsifiability via Social Cycles C_{7C_7}	39
6.22	Falsifiability via Civilisational Cycles C_{8C_8}	40
6.23	Falsifiability via Scientific Cycles C_{9C_9}	40
6.24	Falsifiability via Dynamics and Collapse	40
6.25	Falsifier and collapse boundary	40
6.26	Theorem fate correction after external review	40
6.27	Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks	41
6.28	What would count as a real counterexample	41
6.29	Mathematics: Strict closure and counterexample playbook	41
6.30	Simulation and Counterexample Search	41
6.31	Reviewer challenge 3: Minimality Component Witness	41
6.32	Reviewer challenge 4: Klevel Transition Witness	42
6.33	Falsifiability via Symbolic Systems and Writing	42
6.34	Falsifiability via Knowledge Systems and Epistemic Structure	42
6.35	(P10.11) Completeness Is Structurally Limited	42
6.36	(i) Structural falsifiability	42
6.37	Replay Matrix for Independent Review	43
6.38	Reviewer challenge 5: Numeric Replay Quarantine	43
6.39	Attack Family 3: The Empirical Rows Are Too Weak	43
6.40	Formal-Mathematics Reviewer	43
6.41	Empirical-Statistics Reviewer	43
6.42	Publication-Metadata Reviewer	44
6.43	Worked Attack Transcript B: Replay Surface	44
6.44	Reviewer challenge 7: Prior Art Positioning	44
6.45	Worked Attack Transcript C: Prior-Art Surface	44
6.46	Phenomenon Coverage and Limits	44
6.47	Reviewer challenge 6: Phenomenon Coverage Model Card	45
6.48	Reviewer challenge 8: Attack Matrix Distinct Surface Coverage	45

6.49	Attack Family 4: Phenomenon Coverage Is Inflated	45
6.50	Visual Route - Reviewer Route Figures	45
6.51	Reviewer Navigation Matrix and Audit Checklist	45
6.52	Review questions, compact route, and decisive checks	46
6.53	Audit, provenance, and review posture	46
6.54	Compact reading order by reviewer role	46
6.55	Rapid critical reviewer	46
6.56	Formal mathematical reviewer	46
6.57	Generalist scientific reviewer	47
6.58	Reviewer note	47
6.59	Tuple and hierarchy limits	47
6.60	Reviewer acknowledgement	47
6.61	Hostile-review backlog	47
6.62	Shortest hostile-review checklist	47
6.63	Reviewer Entry Points	48
6.64	Does OC actually explain minimality versus relabeling attack?	48
6.65	Reviewer Checklist	48
6.66	Reviewer Burden and Author Burden	48
6.67	Reviewer Response Method	48
6.68	Claim Boundary and Research Limits	49
6.69	Reviewer challenge 1: Theorem Proof Binding	49
6.70	Reviewer challenge 2: Theorem Public Surface Binding	49
6.71	Adversarial Review Method	49
6.72	Attack Family 1: The Theory Is Only a Reframing	49
6.73	Attack Family 2: The Formal Claims Are Theatre	50
6.74	Attack Family 5: Public Wording Outruns Evidence	50
6.75	Attack Family 6: The Package Is Not a Scientific Text	50
6.76	Reviewer Role Playbook	50
6.77	Computational-Semantics Reviewer	50
6.78	Release-Engineering Reviewer	51
6.79	Synthesis Reviewer	51
6.80	Minimum Hostile Review Walkthrough	51
6.81	End-to-End Attack Drill	51
6.82	Cheap Review Cascade	51
6.83	Worked Attack Transcript A: Theorem Surface	51
6.84	Worked Attack Transcript D: Public Surface	52
6.85	Worked Attack Transcript E: Synthesis Surface	52
6.86	Falsifiability via Institutions and Governance	52
6.87	Limitations of the current formalism	52
6.88	Limits of recovery	52
6.89	Extended Falsifiability and Experimental Programme	53
6.90	Meaning of Falsifiability for Structural Theories	53
6.91	Structural vs phenomenological falsifiability	53
6.92	Structural falsification channels	53
6.93	Cross-domain falsification logic	53
6.94	Falsification criteria and tests	53
6.95	Source-tradition remediation after external review	54
6.96	K-level criteria and falsifiers after external review	54

6.97	Theorem fate correction after external review	54
6.98	Meta-limit extension	54
6.99	Type~VII Experiments: Detectability and Falsification	54
6.100	Relation to Falsifiability	55
6.101	Falsifiability of K_0K_0	55
6.102	Fundamental Falsifiability Principles for K_0K_0	55
6.103	Structural Falsifiability Conditions	55
6.104	Boundary and Region Falsifiability	55
6.105	Falsifiability of K_1K_1	56
6.106	Structural and Topological Falsifiability	56
6.107	Energy Functional and Threshold Falsifiability	56
6.108	Cycle and Evolution Falsifiability	56
6.109	Embedding-Space and Transition Falsifiability	56
6.110	Falsifiability of K_10	56
6.111	Falsifiability via Reflexive and Meta-Theoretical Coherence	57
6.112	Falsifiability via Category-Theoretical and Functorial Structure	57
6.113	Falsifiability via Modal Spaces and Modal Dimensions	57
6.114	Falsifiability via Meta-Level Potentials P_10	57
6.115	Falsifiability via Meta-Flows $J^\wedge(10)$	57
6.116	Falsifiability via Meta-Cycles C_10	58
6.117	Falsifiability via Structural Tension T_10	58
6.118	Falsifiability of the Transition $K_9 K_10$	58
6.119	Falsifiability of K_11	58
6.120	Falsifiability via Meta-Meta Coherence	58
6.121	Falsifiability via Higher-Order Categorical Structures	59
6.122	Falsifiability via Higher-Order Modal Spaces	59
6.123	Falsifiability via Potentials P_11	59
6.124	Falsifiability via Flows $J^\wedge(11)$	59
6.125	Falsifiability via Cycles C_11	59
6.126	Falsifiability via Structural Tension T_11	59
6.127	Falsifiability of the Transition $K_10 K_11$	60
6.128	Falsifiability of K_12	60
6.129	Falsifiability via Universal Structural Consistency	60
6.130	Falsifiability via Universal Modal and Categorical Structure	60
6.131	Falsifiability via Universal Potentials P_12	60
6.132	Falsifiability via Universal Flows $J^\wedge(12)$	61
6.133	Falsifiability via Universal Cycles C_12	61
6.134	Falsifiability via Universal Boundaries (K_12)	61
6.135	Falsifiability of $K_11 K_12$ Transition	61
6.136	Falsifiability of K_2K_2	61
6.137	Topological and Connectivity Falsifiability	62
6.138	Percolation and Threshold Falsifiability	62
6.139	Boundary and Admissibility Falsifiability	62
6.140	Potentials, Flows and Field Falsifiability	62
6.141	Dynamic and Evolutionary Falsifiability	62
6.142	Transition and Embedding Falsifiability	62
6.143	Falsifiability of K_3K_3	63
6.144	Falsifiability via Chemical Configuration	63

6.145	Falsifiability via Reaction Networks	63
6.146	Falsifiability via RAF Networks and Closure	63
6.147	Falsifiability via Chemical Potentials and Thresholds	63
6.148	Falsifiability via Boundaries and Admissibility	64
6.149	Falsifiability via Dynamics and Evolution	64
6.150	Falsifiability via Transition $K_2 K_3 K_2 K_3$	64
6.151	Falsifiability of $K_4 K_4$	64
6.152	Falsifiability via Compartment Formation	64
6.153	Falsifiability via Biological Potentials $P_4 P_4$	64
6.154	Falsifiability via Flows $J_4 J_4$	65
6.155	Falsifiability via Cycles $C_4 C_4$	65
6.156	Falsifiability via Membrane-Bound Information and Regulation	65
6.157	Falsifiability via Structural Tension $T_4 T_4$	65
6.158	Falsifiability via Transition $K_3 K_4 K_3 K_4$	65
6.159	Falsifiability of $K_5 K_5$	66
6.160	Falsifiability via Membrane and Electrical Boundary	66
6.161	Falsifiability via Electrical Potentials $P_5 P_5$	66
6.162	Falsifiability via Ion Flows $J_5 J_5$	66
6.163	Falsifiability via Neural Cycles $C_5 C_5$	66
6.164	Falsifiability via Stochastic Logic and Information Flow	66
6.165	Falsifiability via Structural Tension $T_5 T_5$	67
6.166	Falsifiability of Transition $K_4 K_5 K_4 K_5$	67
6.167	Falsifiability of $K_6 K_6$	67
6.168	Falsifiability via Representational Structure	67
6.169	Falsifiability via Binding Dynamics	67
6.170	Falsifiability via Predictive Processing	68
6.171	Falsifiability via Comparison and Selection	68
6.172	Falsifiability via Memory and Stability	68
6.173	Falsifiability via Cognitive Flows $J_6 J_6$	68
6.174	Falsifiability via Cognitive Cycles $C_6 C_6$	68
6.175	Falsifiability via Structural Tension $T_6 T_6$	68
6.176	Falsifiability of Transition $K_5 K_6 K_5 K_6$	69
6.177	Falsifiability of $K_7 K_7$	69
6.178	Falsifiability via Communication Structure	69
6.179	Falsifiability via Trust and Cohesion	69
6.180	Falsifiability via Roles and Normative Structure	69
6.181	Falsifiability via Coordination Dynamics	70
6.182	Falsifiability via Resource Flows $J_7 J_7$	70
6.183	Falsifiability via Social Cycles $C_7 C_7$	70
6.184	Falsifiability via Structural Tension $T_7 T_7$	70
6.185	Falsifiability of Transition $K_6 K_7 K_6 K_7$	70
6.186	Falsifiability of $K_8 K_8$	70
6.187	Falsifiability via Technology and Infrastructure	71
6.188	Falsifiability via Cultural Dynamics	71
6.189	Falsifiability via Civilisational Flows $J_8 J_8$	71
6.190	Falsifiability via Civilisational Cycles $C_8 C_8$	71
6.191	Falsifiability via Structural Tension $T_8 T_8$	71
6.192	Falsifiability of Transition $K_7 K_8 K_7 K_8$	72

6.193	Falsifiability of K_9K_9	72
6.194	Falsifiability via Coherence and Logical Consistency	72
6.195	Falsifiability via Paradigm Dynamics	72
6.196	Falsifiability via Epistemic Potentials and Explanatory Power	72
6.197	Falsifiability via Methodological Structure	72
6.198	Falsifiability via Scientific Flows J_9J_9	73
6.199	Falsifiability via Scientific Cycles C_9C_9	73
6.200	Falsifiability via Structural Tension T_9T_9	73
6.201	Falsifiability of the Transition K_8 K_9K_8 K_9	73
6.202	Falsifiability	73
6.203	Structural Meaning of Falsifiability	74
6.204	Universal Falsifiability Criteria	74
6.205	Cross-Level Falsifiability Structure	74
6.206	Falsifiability via Dynamics and Collapse	74
6.207	Falsifiability via Embedding Spaces	74
6.208	Falsifiability in Practice: Multi-Domain Implementation	74
6.209	Falsifier and collapse boundary	75
6.210	Claim Boundary and Research Limits	75
6.211	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE	75
6.212	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE	75
6.213	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE	76
6.214	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE	76
6.215	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE	76
6.216	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE	76
6.217	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE	76
6.218	FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE	77
6.219	FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE	77
6.220	FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE	77
6.221	FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE	77
6.222	FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE	78
6.223	FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE	78
6.224	FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE	78
6.225	FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE	78

6.226	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE	78
6.227	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE	79
6.228	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE	79
6.229	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE	79
6.230	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE	79
6.231	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE	80
6.232	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE	80
6.233	FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE	80
6.234	FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE	80
6.235	FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE	80
6.236	FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE	81
6.237	FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE	81
6.238	FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE	81
6.239	FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE	81
6.240	FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE	82
6.241	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE	82
6.242	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE	82
6.243	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE	82
6.244	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE	82
6.245	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE	83
6.246	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE	83
6.247	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE	83
6.248	FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE	83
6.249	FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE	84

6.250	FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE	84
6.251	FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE	84
6.252	FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE	84
6.253	FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE	84
6.254	FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE	85
6.255	FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE	85
6.256	PHYSICAL_REVIEW_RESEARCH	85
6.257	Minimum Independent Scientific Review Packet	85
6.258	Limitations	86
6.259	Journal Owner-Review Packages	86
6.260	Limitations of the current formalism	86
6.261	Extended Falsifiability and Experimental Programme	86
6.262	Meaning of Falsifiability for Structural Theories	86
6.263	Structural vs phenomenological falsifiability	86
6.264	Structural falsification channels	87
6.265	Cross-domain falsification logic	87
6.266	Falsification criteria and tests	87
6.267	Source-tradition remediation after external review	87
6.268	Meta-limit extension	87
6.269	Type~VII Experiments: Detectability and Falsification	88
6.270	Relation to Falsifiability	88
6.271	Falsifiability of K_1K_1	88
6.272	Structural and Topological Falsifiability	88
6.273	Embedding-Space and Transition Falsifiability	88
6.274	Falsifiability of K_10	88
6.275	Falsifiability via Reflexive and Meta-Theoretical Coherence	89
6.276	Falsifiability via Category-Theoretical and Functorial Structure	89
6.277	Falsifiability via Meta-Level Potentials P_10	89
6.278	Falsifiability via Meta-Flows $J^\wedge(10)$	89
6.279	Falsifiability via Structural Tension T_10	89
6.280	Falsifiability of the Transition $K_9 K_10$	90
6.281	Falsifiability of K_11	90
6.282	Falsifiability via Meta-Meta Coherence	90
6.283	Falsifiability via Higher-Order Categorical Structures	90
6.284	Falsifiability via Higher-Order Modal Spaces	90
6.285	Falsifiability via Potentials P_11	91
6.286	Falsifiability via Flows $J^\wedge(11)$	91
6.287	Falsifiability via Structural Tension T_11	91
6.288	Falsifiability of the Transition $K_10 K_11$	91
6.289	Falsifiability of K_12	91
6.290	Falsifiability via Universal Structural Consistency	91
6.291	Falsifiability via Universal Modal and Categorical Structure	92

6.292	Falsifiability via Universal Potentials P_12	92
6.293	Falsifiability via Universal Flows $J^{\wedge}(12)$	92
6.294	Falsifiability via Universal Boundaries (K_12)	92
6.295	Falsifiability of K_11 K_12 Transition	92
6.296	Falsifiability of K_2K_2	93
6.297	Topological and Connectivity Falsifiability	93
6.298	Potentials, Flows and Field Falsifiability	93
6.299	Dynamic and Evolutionary Falsifiability	93
6.300	Transition and Embedding Falsifiability	93
6.301	Falsifiability of K_3K_3	93
6.302	Falsifiability via Chemical Configuration	94
6.303	Falsifiability via Reaction Networks	94
6.304	Falsifiability via RAF Networks and Closure	94
6.305	Falsifiability via Boundaries and Admissibility	94
6.306	Falsifiability via Dynamics and Evolution	94
6.307	Falsifiability via Transition K_2 K_3K_2 K_3	95
6.308	Falsifiability of K_4K_4	95
6.309	Falsifiability via Compartment Formation	95
6.310	Falsifiability via Biological Potentials P_4P_4	95
6.311	Falsifiability via Flows J_4J_4	95
6.312	Falsifiability via Membrane-Bound Information and Regulation	95
6.313	Falsifiability via Structural Tension T_4T_4	96
6.314	Falsifiability via Transition K_3 K_4K_3 K_4	96
6.315	Falsifiability of K_5K_5	96
6.316	Falsifiability via Electrical Potentials P_5P_5	96
6.317	Falsifiability via Ion Flows J_5J_5	96
6.318	Falsifiability via Stochastic Logic and Information Flow	97
6.319	Falsifiability via Structural Tension T_5T_5	97
6.320	Falsifiability of Transition K_4 K_5K_4 K_5	97
6.321	Falsifiability of K_6K_6	97
6.322	Falsifiability via Representational Structure	97
6.323	Falsifiability via Binding Dynamics	97
6.324	Falsifiability via Predictive Processing	98
6.325	Falsifiability via Comparison and Selection	98
6.326	Falsifiability via Memory and Stability	98
6.327	Falsifiability via Cognitive Flows J_6J_6	98
6.328	Falsifiability via Structural Tension T_6T_6	98
6.329	Falsifiability of Transition K_5 K_6K_5 K_6	99
6.330	Falsifiability of K_7K_7	99
6.331	Falsifiability via Communication Structure	99
6.332	Falsifiability via Trust and Cohesion	99
6.333	Falsifiability via Roles and Normative Structure	99
6.334	Falsifiability via Coordination Dynamics	99
6.335	Falsifiability via Resource Flows J_7J_7	100
6.336	Falsifiability via Structural Tension T_7T_7	100
6.337	Falsifiability of Transition K_6 K_7K_6 K_7	100
6.338	Falsifiability of K_8K_8	100
6.339	Falsifiability via Technology and Infrastructure	100

6.340	Falsifiability via Cultural Dynamics	101
6.341	Falsifiability via Civilisational Flows J_8J_8	101
6.342	Falsifiability via Structural Tension T_8T_8	101
6.343	Falsifiability of Transition K_7 K_8K_7 K_8	101
6.344	Falsifiability of K_9K_9	101
6.345	Falsifiability via Coherence and Logical Consistency	101
6.346	Falsifiability via Paradigm Dynamics	102
6.347	Falsifiability via Epistemic Potentials and Explanatory Power	102
6.348	Falsifiability via Methodological Structure	102
6.349	Falsifiability via Scientific Flows J_9J_9	102
6.350	Falsifiability via Structural Tension T_9T_9	102
6.351	Falsifiability of the Transition K_8 K_9K_8 K_9	103
6.352	Falsifiability	103
6.353	Structural Meaning of Falsifiability	103
6.354	Universal Falsifiability Criteria	103
6.355	Cross-Level Falsifiability Structure	103
6.356	Falsifiability via Embedding Spaces	103
6.357	Falsifiability in Practice: Multi-Domain Implementation	104
7	Source Trace	104

1 Reader Orientation and Claim Boundaries

1.1 Reader Contract

Reader Contract is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Reader Contract. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Reader Contract to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Reader Contract states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Reader Contract synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

1.2 Intended Audiences and Reading Paths

Intended Audiences and Reading Paths is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Intended Audiences and Reading Paths. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Intended Audiences and Reading Paths to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited

proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Intended Audiences and Reading Paths states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Intended Audiences and Reading Paths synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

1.3 What OC Core Claims

What OC Core Claims is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What OC Core Claims. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

What OC Core Claims to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

What OC Core Claims states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

What OC Core Claims synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

1.4 What OC Core Does Not Claim

What OC Core Does Not Claim is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for What OC Core Does Not Claim. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

What OC Core Does Not Claim to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

What OC Core Does Not Claim states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

What OC Core Does Not Claim synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

1.5 Release Scope vs Full Science Program

Release Scope vs Full Science Program is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Release Scope vs Full Science Program. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Release Scope vs Full Science Program to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only

becomes reviewable when its boundary, falsifier, and negative side are visible.

Release Scope vs Full Science Program states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Release Scope vs Full Science Program synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

1.6 Claim Promotion, Demotion, and Background Obligations

Claim Promotion, Demotion, and Background Obligations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Claim Promotion, Demotion, and Background Obligations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Claim Promotion, Demotion, and Background Obligations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Claim Promotion, Demotion, and Background Obligations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Claim Promotion, Demotion, and Background Obligations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next

obligation begins by defining the next object before asking the reader to accept claims about it.

2 Falsifiers and Negative Controls

2.1 Falsifier Registry

Falsifier Registry is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Falsifier Registry. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Falsifier Registry to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Falsifier Registry states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Falsifier Registry synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2 Counterexample Search

Counterexample Search is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Counterexample Search. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Counterexample Search to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics,

replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Counterexample Search states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Counterexample Search synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3 Known Failure Modes

Known Failure Modes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Known Failure Modes. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Known Failure Modes to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Known Failure Modes states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Known Failure Modes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength

beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4 Unsupported Strong Claims

Unsupported Strong Claims is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Unsupported Strong Claims. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Unsupported Strong Claims to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Unsupported Strong Claims states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Unsupported Strong Claims synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5 Demotion Rules

Demotion Rules is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Demotion Rules. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Demotion Rules to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computa-

tional evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Demotion Rules states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Demotion Rules synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6 Residual Scientific Risks

Residual Scientific Risks is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Residual Scientific Risks. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Residual Scientific Risks to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Residual Scientific Risks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Residual Scientific Risks synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize

established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.7 Full-Science Background Obligations

Full-Science Background Obligations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Full-Science Background Obligations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Full-Science Background Obligations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Full-Science Background Obligations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Full-Science Background Obligations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

3 Novelty and Prior-Art Boundary

3.1 Novelty Criteria

Novelty Criteria is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Novelty Criteria. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Novelty Criteria to proof, evidence, or replay carries the evidential burden for the surrounding

claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Novelty Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Novelty Criteria synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

3.2 Non-Equivalence Criteria

Non-Equivalence Criteria is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Non-Equivalence Criteria. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Non-Equivalence Criteria to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Non-Equivalence Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Non-Equivalence Criteria synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on

reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

3.3 Overlap With Prior Art

Overlap With Prior Art is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Overlap With Prior Art. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Overlap With Prior Art to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Overlap With Prior Art states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Overlap With Prior Art synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

3.4 Residual-Delta Analysis

Residual-Delta Analysis is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Residual-Delta Analysis. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Residual-Delta Analysis to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite

semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Residual-Delta Analysis states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Residual-Delta Analysis synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

3.5 Competitor-by-Competitor Comparison

Competitor-by-Competitor Comparison is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Competitor-by-Competitor Comparison. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Competitor-by-Competitor Comparison to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Competitor-by-Competitor Comparison states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Competitor-by-Competitor Comparison synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws

on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

3.6 Response to “This Is Just Another Theory”

Response to “This Is Just Another Theory” is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Response to “This Is Just Another Theory”. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Response to “This Is Just Another Theory” to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Response to “This Is Just Another Theory” states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Response to “This Is Just Another Theory” synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

3.7 Positioning of OC Core

Positioning of OC Core is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Positioning of OC Core. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Positioning of OC Core to proof, evidence, or replay carries the evidential burden for the sur-

rounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Positioning of OC Core states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Positioning of OC Core synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4 Reviewer Protocol and Attack Surface

4.1 Review Protocol

Review Protocol is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Review Protocol. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Review Protocol to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Review Protocol states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Review Protocol synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4.2 Cerberus Role Map

Cerberus Role Map is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Cerberus Role Map. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Cerberus Role Map to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Cerberus Role Map states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Cerberus Role Map synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4.3 Critical and High Findings

Critical and High Findings is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Critical and High Findings. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Critical and High Findings to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Critical and High Findings states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Critical and High Findings synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4.4 Closure Evidence

Closure Evidence is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Closure Evidence. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Closure Evidence to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Closure Evidence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Closure Evidence synthesizes the local route for the reader. It states what has been established,

what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4.5 Claim Corrections

Claim Corrections is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Claim Corrections. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Claim Corrections to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Claim Corrections states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Claim Corrections synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4.6 Hostile Reviewer Objections

Hostile Reviewer Objections is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Hostile Reviewer Objections. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Hostile Reviewer Objections to proof, evidence, or replay carries the evidential burden for the

surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Hostile Reviewer Objections states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Hostile Reviewer Objections synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4.7 Response Matrix

Response Matrix is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Response Matrix. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Response Matrix to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Response Matrix states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Response Matrix synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility

governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

4.8 Remaining Non-Blocking Caveats

Remaining Non-Blocking Caveats is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Remaining Non-Blocking Caveats. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Remaining Non-Blocking Caveats to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Remaining Non-Blocking Caveats states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Remaining Non-Blocking Caveats synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5 Versioning, Citation, and Release Identity

5.1 Versioning and Release Identity

Versioning and Release Identity is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Versioning and Release Identity. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Versioning and Release Identity to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Versioning and Release Identity states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Versioning and Release Identity synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5.2 Public Release Asset Set

Public Release Asset Set is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Public Release Asset Set. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Public Release Asset Set to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Public Release Asset Set states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Public Release Asset Set synthesizes the local route for the reader. It states what has been es-

tablished, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5.3 Journal Package Map

Journal Package Map is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Journal Package Map. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Journal Package Map to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Journal Package Map states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Journal Package Map synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5.4 Citation and Metadata Governance

Citation and Metadata Governance is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Citation and Metadata Governance. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Citation and Metadata Governance to proof, evidence, or replay carries the evidential burden for

the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Citation and Metadata Governance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Citation and Metadata Governance synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5.5 Publication Boundary

Publication Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Publication Boundary. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Publication Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Publication Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Publication Boundary synthesizes the local route for the reader. It states what has been established,

what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5.6 Owner Approval Boundary

Owner Approval Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Owner Approval Boundary. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Owner Approval Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Owner Approval Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Owner Approval Boundary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5.7 Incident and Known-Error Lessons

Incident and Known-Error Lessons is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Incident and Known-Error Lessons. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Incident and Known-Error Lessons to proof, evidence, or replay carries the evidential burden for

the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

Incident and Known-Error Lessons states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Incident and Known-Error Lessons synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

5.8 External Use Guidance

External Use Guidance is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for External Use Guidance. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

External Use Guidance to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

External Use Guidance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

External Use Guidance synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on

reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

6 Document Boundary

6.1 Adversarial Review Structure and Audience

Adversarial Review Structure and Audience states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.2 Limits of recovery

Limits of recovery states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.3 K-level criteria and falsifiers after external review

K-level criteria and falsifiers after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.4 Falsifiability of K_0K_0

Falsifiability of K_0K_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.5 Fundamental Falsifiability Principles for K_0K_0

Fundamental Falsifiability Principles for K_0K_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure

and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.6 Structural Falsifiability Conditions

Structural Falsifiability Conditions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.7 Boundary and Region Falsifiability

Boundary and Region Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.8 Energy Functional and Threshold Falsifiability

Energy Functional and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.9 Cycle and Evolution Falsifiability

Cycle and Evolution Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.10 Falsifiability via Modal Spaces and Modal Dimensions

Falsifiability via Modal Spaces and Modal Dimensions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.11 Falsifiability via Meta-Cycles C_10

Falsifiability via Meta-Cycles C_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.12 Falsifiability via Cycles C_11

Falsifiability via Cycles C_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.13 Falsifiability via Universal Cycles C_12

Falsifiability via Universal Cycles C_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.14 Percolation and Threshold Falsifiability

Percolation and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.15 Boundary and Admissibility Falsifiability

Boundary and Admissibility Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.16 Falsifiability via Chemical Potentials and Thresholds

Falsifiability via Chemical Potentials and Thresholds states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.17 Falsifiability via Cycles C_4C_4

Falsifiability via Cycles C_4C_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.18 Falsifiability via Membrane and Electrical Boundary

Falsifiability via Membrane and Electrical Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.19 Falsifiability via Neural Cycles C_5C_5

Falsifiability via Neural Cycles C_5C_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.20 Falsifiability via Cognitive Cycles C_6C_6

Falsifiability via Cognitive Cycles C_6C_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.21 Falsifiability via Social Cycles C_7C_7

Falsifiability via Social Cycles C_7C_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.22 Falsifiability via Civilisational Cycles C_8C_8

Falsifiability via Civilisational Cycles C_8C_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.23 Falsifiability via Scientific Cycles C_9C_9

Falsifiability via Scientific Cycles C_9C_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.24 Falsifiability via Dynamics and Collapse

Falsifiability via Dynamics and Collapse states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.25 Falsifier and collapse boundary

Falsifier and collapse boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.26 Theorem fate correction after external review

Theorem fate correction after external review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.27 Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks

Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.28 What would count as a real counterexample

What would count as a real counterexample carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.29 Mathematics: Strict closure and counterexample playbook

Mathematics: Strict closure and counterexample playbook carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.30 Simulation and Counterexample Search

Simulation and Counterexample Search carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.31 Reviewer challenge 3: Minimality Component Witness

Reviewer challenge 3: Minimality Component Witness carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable.

The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.32 Reviewer challenge 4: Klevel Transition Witness

Reviewer challenge 4: Klevel Transition Witness carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.33 Falsifiability via Symbolic Systems and Writing

Falsifiability via Symbolic Systems and Writing carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.34 Falsifiability via Knowledge Systems and Epistemic Structure

Falsifiability via Knowledge Systems and Epistemic Structure carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.35 (P10.11) Completeness Is Structurally Limited

(P10.11) Completeness Is Structurally Limited carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.36 (i) Structural falsifiability

- (i) Structural falsifiability carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact

paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.37 Replay Matrix for Independent Review

Replay Matrix for Independent Review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.38 Reviewer challenge 5: Numeric Replay Quarantine

Reviewer challenge 5: Numeric Replay Quarantine carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.39 Attack Family 3: The Empirical Rows Are Too Weak

Attack Family 3: The Empirical Rows Are Too Weak carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.40 Formal-Mathematics Reviewer

Formal-Mathematics Reviewer carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.41 Empirical-Statistics Reviewer

Empirical-Statistics Reviewer carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, com-

parator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.42 Publication-Metadata Reviewer

Publication-Metadata Reviewer carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.43 Worked Attack Transcript B: Replay Surface

Worked Attack Transcript B: Replay Surface carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.44 Reviewer challenge 7: Prior Art Positioning

Reviewer challenge 7: Prior Art Positioning states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.45 Worked Attack Transcript C: Prior-Art Surface

Worked Attack Transcript C: Prior-Art Surface states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.46 Phenomenon Coverage and Limits

Phenomenon Coverage and Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical

toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.47 Reviewer challenge 6: Phenomenon Coverage Model Card

Reviewer challenge 6: Phenomenon Coverage Model Card states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.48 Reviewer challenge 8: Attack Matrix Distinct Surface Coverage

Reviewer challenge 8: Attack Matrix Distinct Surface Coverage states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.49 Attack Family 4: Phenomenon Coverage Is Inflated

Attack Family 4: Phenomenon Coverage Is Inflated states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.50 Visual Route - Reviewer Route Figures

Visual Route - Reviewer Route Figures states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.51 Reviewer Navigation Matrix and Audit Checklist

Reviewer Navigation Matrix and Audit Checklist states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure

and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.52 Review questions, compact route, and decisive checks

Review questions, compact route, and decisive checks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.53 Audit, provenance, and review posture

Audit, provenance, and review posture states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.54 Compact reading order by reviewer role

Compact reading order by reviewer role states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.55 Rapid critical reviewer

Rapid critical reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.56 Formal mathematical reviewer

Formal mathematical reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.57 Generalist scientific reviewer

Generalist scientific reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.58 Reviewer note

Reviewer note states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.59 Tuple and hierarchy limits

Tuple and hierarchy limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.60 Reviewer acknowledgement

Reviewer acknowledgement states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.61 Hostile-review backlog

Hostile-review backlog states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.62 Shortest hostile-review checklist

Shortest hostile-review checklist states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical

toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.63 Reviewer Entry Points

Reviewer Entry Points states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.64 Does OC actually explain minimality versus relabeling attack?

Does OC actually explain minimality versus relabeling attack? states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.65 Reviewer Checklist

Reviewer Checklist states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.66 Reviewer Burden and Author Burden

Reviewer Burden and Author Burden states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.67 Reviewer Response Method

Reviewer Response Method states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.68 Claim Boundary and Research Limits

Claim Boundary and Research Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.69 Reviewer challenge 1: Theorem Proof Binding

Reviewer challenge 1: Theorem Proof Binding carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.70 Reviewer challenge 2: Theorem Public Surface Binding

Reviewer challenge 2: Theorem Public Surface Binding carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

6.71 Adversarial Review Method

Adversarial Review Method states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.72 Attack Family 1: The Theory Is Only a Reframing

Attack Family 1: The Theory Is Only a Reframing states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.73 Attack Family 2: The Formal Claims Are Theatre

Attack Family 2: The Formal Claims Are Theatre states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.74 Attack Family 5: Public Wording Outruns Evidence

Attack Family 5: Public Wording Outruns Evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

6.75 Attack Family 6: The Package Is Not a Scientific Text

Attack Family 6: The Package Is Not a Scientific Text states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.76 Reviewer Role Playbook

Reviewer Role Playbook states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.77 Computational-Semantics Reviewer

Computational-Semantics Reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.78 Release-Engineering Reviewer

Release-Engineering Reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.79 Synthesis Reviewer

Synthesis Reviewer states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.80 Minimum Hostile Review Walkthrough

Minimum Hostile Review Walkthrough states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.81 End-to-End Attack Drill

End-to-End Attack Drill states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.82 Cheap Review Cascade

Cheap Review Cascade states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.83 Worked Attack Transcript A: Theorem Surface

Worked Attack Transcript A: Theorem Surface carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery;

exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

6.84 Worked Attack Transcript D: Public Surface

Worked Attack Transcript D: Public Surface states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.85 Worked Attack Transcript E: Synthesis Surface

Worked Attack Transcript E: Synthesis Surface states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.86 Falsifiability via Institutions and Governance

Falsifiability via Institutions and Governance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.87 Limitations of the current formalism

Limitations of the current formalism states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.88 Limits of recovery

Limits of recovery states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.89 Extended Falsifiability and Experimental Programme

Extended Falsifiability and Experimental Programme states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.90 Meaning of Falsifiability for Structural Theories

Meaning of Falsifiability for Structural Theories states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.91 Structural vs phenomenological falsifiability

Structural vs phenomenological falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.92 Structural falsification channels

Structural falsification channels states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.93 Cross-domain falsification logic

Cross-domain falsification logic states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.94 Falsification criteria and tests

Falsification criteria and tests states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It

must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.95 Source-tradition remediation after external review

Source-tradition remediation after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.96 K-level criteria and falsifiers after external review

K-level criteria and falsifiers after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.97 Theorem fate correction after external review

Theorem fate correction after external review carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.98 Meta-limit extension

Meta-limit extension states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.99 Type~VII Experiments: Detectability and Falsification

Type~VII Experiments: Detectability and Falsification states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence.

Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.100 Relation to Falsifiability

Relation to Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.101 Falsifiability of K_0K_0

Falsifiability of K_0K_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.102 Fundamental Falsifiability Principles for K_0K_0

Fundamental Falsifiability Principles for K_0K_0 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.103 Structural Falsifiability Conditions

Structural Falsifiability Conditions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.104 Boundary and Region Falsifiability

Boundary and Region Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.105 Falsifiability of K_1K_1

Falsifiability of K_1K_1 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.106 Structural and Topological Falsifiability

Structural and Topological Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.107 Energy Functional and Threshold Falsifiability

Energy Functional and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.108 Cycle and Evolution Falsifiability

Cycle and Evolution Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.109 Embedding-Space and Transition Falsifiability

Embedding-Space and Transition Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.110 Falsifiability of K_10

Falsifiability of K_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc

recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.111 Falsifiability via Reflexive and Meta-Theoretical Coherence

Falsifiability via Reflexive and Meta-Theoretical Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.112 Falsifiability via Category-Theoretical and Functorial Structure

Falsifiability via Category-Theoretical and Functorial Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.113 Falsifiability via Modal Spaces and Modal Dimensions

Falsifiability via Modal Spaces and Modal Dimensions states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.114 Falsifiability via Meta-Level Potentials P_{10}

Falsifiability via Meta-Level Potentials P_{10} states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.115 Falsifiability via Meta-Flows $J^{\wedge}(10)$

Falsifiability via Meta-Flows $J^{\wedge}(10)$ states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and

source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.116 Falsifiability via Meta-Cycles C_10

Falsifiability via Meta-Cycles C_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.117 Falsifiability via Structural Tension T_10

Falsifiability via Structural Tension T_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.118 Falsifiability of the Transition K_9 K_10

Falsifiability of the Transition K_9 K_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.119 Falsifiability of K_11

Falsifiability of K_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.120 Falsifiability via Meta-Meta Coherence

Falsifiability via Meta-Meta Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.121 Falsifiability via Higher-Order Categorical Structures

Falsifiability via Higher-Order Categorical Structures states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.122 Falsifiability via Higher-Order Modal Spaces

Falsifiability via Higher-Order Modal Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.123 Falsifiability via Potentials P_{11}

Falsifiability via Potentials P_{11} states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.124 Falsifiability via Flows $J^{\wedge}(11)$

Falsifiability via Flows $J^{\wedge}(11)$ states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.125 Falsifiability via Cycles C_{11}

Falsifiability via Cycles C_{11} states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.126 Falsifiability via Structural Tension T_{11}

Falsifiability via Structural Tension T_{11} states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.127 Falsifiability of the Transition K_10 K_11

Falsifiability of the Transition K_10 K_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.128 Falsifiability of K_12

Falsifiability of K_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.129 Falsifiability via Universal Structural Consistency

Falsifiability via Universal Structural Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.130 Falsifiability via Universal Modal and Categorical Structure

Falsifiability via Universal Modal and Categorical Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.131 Falsifiability via Universal Potentials P_12

Falsifiability via Universal Potentials P_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure

and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.132 Falsifiability via Universal Flows $J^{\wedge}(12)$

Falsifiability via Universal Flows $J^{\wedge}(12)$ states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.133 Falsifiability via Universal Cycles C_{12}

Falsifiability via Universal Cycles C_{12} states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.134 Falsifiability via Universal Boundaries (K_{12})

Falsifiability via Universal Boundaries (K_{12}) states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.135 Falsifiability of $K_{11} K_{12}$ Transition

Falsifiability of $K_{11} K_{12}$ Transition states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.136 Falsifiability of K_{2K_2}

Falsifiability of K_{2K_2} states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.137 Topological and Connectivity Falsifiability

Topological and Connectivity Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.138 Percolation and Threshold Falsifiability

Percolation and Threshold Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.139 Boundary and Admissibility Falsifiability

Boundary and Admissibility Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.140 Potentials, Flows and Field Falsifiability

Potentials, Flows and Field Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.141 Dynamic and Evolutionary Falsifiability

Dynamic and Evolutionary Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.142 Transition and Embedding Falsifiability

Transition and Embedding Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.143 Falsifiability of K_3K_3

Falsifiability of K_3K_3 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.144 Falsifiability via Chemical Configuration

Falsifiability via Chemical Configuration states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.145 Falsifiability via Reaction Networks

Falsifiability via Reaction Networks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.146 Falsifiability via RAF Networks and Closure

Falsifiability via RAF Networks and Closure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.147 Falsifiability via Chemical Potentials and Thresholds

Falsifiability via Chemical Potentials and Thresholds states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.148 Falsifiability via Boundaries and Admissibility

Falsifiability via Boundaries and Admissibility states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.149 Falsifiability via Dynamics and Evolution

Falsifiability via Dynamics and Evolution states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.150 Falsifiability via Transition K_2 K_3K_2 K_3

Falsifiability via Transition K_2 K_3K_2 K_3 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.151 Falsifiability of K_4K_4

Falsifiability of K_4K_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.152 Falsifiability via Compartment Formation

Falsifiability via Compartment Formation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.153 Falsifiability via Biological Potentials P_4P_4

Falsifiability via Biological Potentials P_4P_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.154 Falsifiability via Flows J_4J_4

Falsifiability via Flows J_4J_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.155 Falsifiability via Cycles C_4C_4

Falsifiability via Cycles C_4C_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.156 Falsifiability via Membrane-Bound Information and Regulation

Falsifiability via Membrane-Bound Information and Regulation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.157 Falsifiability via Structural Tension T_4T_4

Falsifiability via Structural Tension T_4T_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.158 Falsifiability via Transition K_3 K_4K_3 K_4

Falsifiability via Transition K_3 K_4K_3 K_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.159 Falsifiability of K_5K_5

Falsifiability of K_5K_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.160 Falsifiability via Membrane and Electrical Boundary

Falsifiability via Membrane and Electrical Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.161 Falsifiability via Electrical Potentials P_5P_5

Falsifiability via Electrical Potentials P_5P_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.162 Falsifiability via Ion Flows J_5J_5

Falsifiability via Ion Flows J_5J_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.163 Falsifiability via Neural Cycles C_5C_5

Falsifiability via Neural Cycles C_5C_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.164 Falsifiability via Stochastic Logic and Information Flow

Falsifiability via Stochastic Logic and Information Flow states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or

residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.165 Falsifiability via Structural Tension T_5T_5

Falsifiability via Structural Tension T_5T_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.166 Falsifiability of Transition K_4 K_5K_4 K_5

Falsifiability of Transition K_4 K_5K_4 K_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.167 Falsifiability of K_6K_6

Falsifiability of K_6K_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.168 Falsifiability via Representational Structure

Falsifiability via Representational Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.169 Falsifiability via Binding Dynamics

Falsifiability via Binding Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.170 Falsifiability via Predictive Processing

Falsifiability via Predictive Processing states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.171 Falsifiability via Comparison and Selection

Falsifiability via Comparison and Selection states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.172 Falsifiability via Memory and Stability

Falsifiability via Memory and Stability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.173 Falsifiability via Cognitive Flows J_6J_6

Falsifiability via Cognitive Flows J_6J_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.174 Falsifiability via Cognitive Cycles C_6C_6

Falsifiability via Cognitive Cycles C_6C_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.175 Falsifiability via Structural Tension T_6T_6

Falsifiability via Structural Tension T_6T_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.176 Falsifiability of Transition K_5 K_6K_5 K_6

Falsifiability of Transition K_5 K_6K_5 K_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.177 Falsifiability of K_7K_7

Falsifiability of K_7K_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.178 Falsifiability via Communication Structure

Falsifiability via Communication Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.179 Falsifiability via Trust and Cohesion

Falsifiability via Trust and Cohesion states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.180 Falsifiability via Roles and Normative Structure

Falsifiability via Roles and Normative Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.181 Falsifiability via Coordination Dynamics

Falsifiability via Coordination Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.182 Falsifiability via Resource Flows J_7J_7

Falsifiability via Resource Flows J_7J_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.183 Falsifiability via Social Cycles C_7C_7

Falsifiability via Social Cycles C_7C_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.184 Falsifiability via Structural Tension T_7T_7

Falsifiability via Structural Tension T_7T_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.185 Falsifiability of Transition K_6 K_7K_6 K_7

Falsifiability of Transition K_6 K_7K_6 K_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.186 Falsifiability of K_8K_8

Falsifiability of K_8K_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical

toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.187 Falsifiability via Technology and Infrastructure

Falsifiability via Technology and Infrastructure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.188 Falsifiability via Cultural Dynamics

Falsifiability via Cultural Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.189 Falsifiability via Civilisational Flows J_8J_8

Falsifiability via Civilisational Flows J_8J_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.190 Falsifiability via Civilisational Cycles C_8C_8

Falsifiability via Civilisational Cycles C_8C_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.191 Falsifiability via Structural Tension T_8T_8

Falsifiability via Structural Tension T_8T_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.192 Falsifiability of Transition K_7 K_8K_7 K_8

Falsifiability of Transition K_7 K_8K_7 K_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.193 Falsifiability of K_9K_9

Falsifiability of K_9K_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.194 Falsifiability via Coherence and Logical Consistency

Falsifiability via Coherence and Logical Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.195 Falsifiability via Paradigm Dynamics

Falsifiability via Paradigm Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.196 Falsifiability via Epistemic Potentials and Explanatory Power

Falsifiability via Epistemic Potentials and Explanatory Power states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.197 Falsifiability via Methodological Structure

Falsifiability via Methodological Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.198 Falsifiability via Scientific Flows J_9J_9

Falsifiability via Scientific Flows J_9J_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.199 Falsifiability via Scientific Cycles C_9C_9

Falsifiability via Scientific Cycles C_9C_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.200 Falsifiability via Structural Tension T_9T_9

Falsifiability via Structural Tension T_9T_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.201 Falsifiability of the Transition K_8 K_9K_8 K_9

Falsifiability of the Transition K_8 K_9K_8 K_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.202 Falsifiability

Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.203 Structural Meaning of Falsifiability

Structural Meaning of Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.204 Universal Falsifiability Criteria

Universal Falsifiability Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.205 Cross-Level Falsifiability Structure

Cross-Level Falsifiability Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.206 Falsifiability via Dynamics and Collapse

Falsifiability via Dynamics and Collapse states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.207 Falsifiability via Embedding Spaces

Falsifiability via Embedding Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.208 Falsifiability in Practice: Multi-Domain Implementation

Falsifiability in Practice: Multi-Domain Implementation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or

residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.209 Falsifier and collapse boundary

Falsifier and collapse boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.210 Claim Boundary and Research Limits

Claim Boundary and Research Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.211 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.212 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.213 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.214 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.215 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.216 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.217 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the

presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.218 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.219 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.220 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.221 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.222 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.223 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.224 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.225 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.226 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the

presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.227 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.228 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.229 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.230 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.231 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.232 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.233 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.234 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.235 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the

presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.236 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.237 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.238 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.239 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.240 FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-RESIDUE-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.241 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.242 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.243 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.244 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the

presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.245 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.246 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.247 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.248 FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-REBIRTH-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.249 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.250 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.251 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.252 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.253 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the

presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.254 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.255 FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE

FM-T133-ID-TT-REBIRTH-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.256 PHYSICAL_REVIEW_RESEARCH

PHYSICAL_REVIEW_RESEARCH states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.257 Minimum Independent Scientific Review Packet

Minimum Independent Scientific Review Packet states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.258 Limitations

Limitations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

6.259 Journal Owner-Review Packages

Journal Owner-Review Packages states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.260 Limitations of the current formalism

Limitations of the current formalism states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.261 Extended Falsifiability and Experimental Programme

Extended Falsifiability and Experimental Programme states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.262 Meaning of Falsifiability for Structural Theories

Meaning of Falsifiability for Structural Theories states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.263 Structural vs phenomenological falsifiability

Structural vs phenomenological falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to

overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.264 Structural falsification channels

Structural falsification channels states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.265 Cross-domain falsification logic

Cross-domain falsification logic states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.266 Falsification criteria and tests

Falsification criteria and tests states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.267 Source-tradition remediation after external review

Source-tradition remediation after external review states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.268 Meta-limit extension

Meta-limit extension states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.269 Type~VII Experiments: Detectability and Falsification

Type~VII Experiments: Detectability and Falsification states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.270 Relation to Falsifiability

Relation to Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.271 Falsifiability of K_1K_1

Falsifiability of K_1K_1 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.272 Structural and Topological Falsifiability

Structural and Topological Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.273 Embedding-Space and Transition Falsifiability

Embedding-Space and Transition Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.274 Falsifiability of K_10

Falsifiability of K_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must

name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.275 Falsifiability via Reflexive and Meta-Theoretical Coherence

Falsifiability via Reflexive and Meta-Theoretical Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.276 Falsifiability via Category-Theoretical and Functorial Structure

Falsifiability via Category-Theoretical and Functorial Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.277 Falsifiability via Meta-Level Potentials P₁₀

Falsifiability via Meta-Level Potentials P₁₀ states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.278 Falsifiability via Meta-Flows J^{^(10)}

Falsifiability via Meta-Flows J^{^(10)} states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.279 Falsifiability via Structural Tension T₁₀

Falsifiability via Structural Tension T₁₀ states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure

and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.280 Falsifiability of the Transition K_9 K_10

Falsifiability of the Transition K_9 K_10 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.281 Falsifiability of K_11

Falsifiability of K_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.282 Falsifiability via Meta-Meta Coherence

Falsifiability via Meta-Meta Coherence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.283 Falsifiability via Higher-Order Categorical Structures

Falsifiability via Higher-Order Categorical Structures states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.284 Falsifiability via Higher-Order Modal Spaces

Falsifiability via Higher-Order Modal Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.285 Falsifiability via Potentials P_11

Falsifiability via Potentials P_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.286 Falsifiability via Flows $\hat{J}(11)$

Falsifiability via Flows $\hat{J}(11)$ states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.287 Falsifiability via Structural Tension T_11

Falsifiability via Structural Tension T_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.288 Falsifiability of the Transition K_10 K_11

Falsifiability of the Transition K_10 K_11 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.289 Falsifiability of K_12

Falsifiability of K_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.290 Falsifiability via Universal Structural Consistency

Falsifiability via Universal Structural Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.291 Falsifiability via Universal Modal and Categorical Structure

Falsifiability via Universal Modal and Categorical Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.292 Falsifiability via Universal Potentials P_12

Falsifiability via Universal Potentials P_12 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.293 Falsifiability via Universal Flows $J^{\wedge}(12)$

Falsifiability via Universal Flows $J^{\wedge}(12)$ states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.294 Falsifiability via Universal Boundaries (K_12)

Falsifiability via Universal Boundaries (K_12) states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.295 Falsifiability of K_11 K_12 Transition

Falsifiability of K_11 K_12 Transition states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.296 Falsifiability of K_2K_2

Falsifiability of K_2K_2 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.297 Topological and Connectivity Falsifiability

Topological and Connectivity Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.298 Potentials, Flows and Field Falsifiability

Potentials, Flows and Field Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.299 Dynamic and Evolutionary Falsifiability

Dynamic and Evolutionary Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.300 Transition and Embedding Falsifiability

Transition and Embedding Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.301 Falsifiability of K_3K_3

Falsifiability of K_3K_3 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical

toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.302 Falsifiability via Chemical Configuration

Falsifiability via Chemical Configuration states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.303 Falsifiability via Reaction Networks

Falsifiability via Reaction Networks states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.304 Falsifiability via RAF Networks and Closure

Falsifiability via RAF Networks and Closure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.305 Falsifiability via Boundaries and Admissibility

Falsifiability via Boundaries and Admissibility states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.306 Falsifiability via Dynamics and Evolution

Falsifiability via Dynamics and Evolution states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.307 Falsifiability via Transition K_2 K_3K_2 K_3

Falsifiability via Transition K_2 K_3K_2 K_3 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.308 Falsifiability of K_4K_4

Falsifiability of K_4K_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.309 Falsifiability via Compartment Formation

Falsifiability via Compartment Formation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.310 Falsifiability via Biological Potentials P_4P_4

Falsifiability via Biological Potentials P_4P_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.311 Falsifiability via Flows J_4J_4

Falsifiability via Flows J_4J_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.312 Falsifiability via Membrane-Bound Information and Regulation

Falsifiability via Membrane-Bound Information and Regulation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection,

or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.313 Falsifiability via Structural Tension T_4T_4

Falsifiability via Structural Tension T_4T_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.314 Falsifiability via Transition K_3 K_4K_3 K_4

Falsifiability via Transition K_3 K_4K_3 K_4 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.315 Falsifiability of K_5K_5

Falsifiability of K_5K_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.316 Falsifiability via Electrical Potentials P_5P_5

Falsifiability via Electrical Potentials P_5P_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.317 Falsifiability via Ion Flows J_5J_5

Falsifiability via Ion Flows J_5J_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.318 Falsifiability via Stochastic Logic and Information Flow

Falsifiability via Stochastic Logic and Information Flow states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.319 Falsifiability via Structural Tension T_5T_5

Falsifiability via Structural Tension T_5T_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.320 Falsifiability of Transition K_4 K_5K_4 K_5

Falsifiability of Transition K_4 K_5K_4 K_5 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.321 Falsifiability of K_6K_6

Falsifiability of K_6K_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.322 Falsifiability via Representational Structure

Falsifiability via Representational Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.323 Falsifiability via Binding Dynamics

Falsifiability via Binding Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate

scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.324 Falsifiability via Predictive Processing

Falsifiability via Predictive Processing states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.325 Falsifiability via Comparison and Selection

Falsifiability via Comparison and Selection states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.326 Falsifiability via Memory and Stability

Falsifiability via Memory and Stability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.327 Falsifiability via Cognitive Flows J_6J_6

Falsifiability via Cognitive Flows J_6J_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.328 Falsifiability via Structural Tension T_6T_6

Falsifiability via Structural Tension T_6T_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.329 Falsifiability of Transition K_5 K_6K_5 K_6

Falsifiability of Transition K_5 K_6K_5 K_6 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.330 Falsifiability of K_7K_7

Falsifiability of K_7K_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.331 Falsifiability via Communication Structure

Falsifiability via Communication Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.332 Falsifiability via Trust and Cohesion

Falsifiability via Trust and Cohesion states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.333 Falsifiability via Roles and Normative Structure

Falsifiability via Roles and Normative Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.334 Falsifiability via Coordination Dynamics

Falsifiability via Coordination Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.335 Falsifiability via Resource Flows J_7J_7

Falsifiability via Resource Flows J_7J_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.336 Falsifiability via Structural Tension T_7T_7

Falsifiability via Structural Tension T_7T_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.337 Falsifiability of Transition K_6 K_7K_6 K_7

Falsifiability of Transition K_6 K_7K_6 K_7 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.338 Falsifiability of K_8K_8

Falsifiability of K_8K_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.339 Falsifiability via Technology and Infrastructure

Falsifiability via Technology and Infrastructure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.340 Falsifiability via Cultural Dynamics

Falsifiability via Cultural Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.341 Falsifiability via Civilisational Flows J_8J_8

Falsifiability via Civilisational Flows J_8J_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.342 Falsifiability via Structural Tension T_8T_8

Falsifiability via Structural Tension T_8T_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.343 Falsifiability of Transition K_7 K_8K_7 K_8

Falsifiability of Transition K_7 K_8K_7 K_8 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.344 Falsifiability of K_9K_9

Falsifiability of K_9K_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.345 Falsifiability via Coherence and Logical Consistency

Falsifiability via Coherence and Logical Consistency states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk

carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.346 Falsifiability via Paradigm Dynamics

Falsifiability via Paradigm Dynamics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.347 Falsifiability via Epistemic Potentials and Explanatory Power

Falsifiability via Epistemic Potentials and Explanatory Power states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.348 Falsifiability via Methodological Structure

Falsifiability via Methodological Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.349 Falsifiability via Scientific Flows J_9J_9

Falsifiability via Scientific Flows J_9J_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.350 Falsifiability via Structural Tension T_9T_9

Falsifiability via Structural Tension T_9T_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.351 Falsifiability of the Transition K_8 K_9K_8 K_9

Falsifiability of the Transition K_8 K_9K_8 K_9 states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.352 Falsifiability

Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.353 Structural Meaning of Falsifiability

Structural Meaning of Falsifiability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.354 Universal Falsifiability Criteria

Universal Falsifiability Criteria states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.355 Cross-Level Falsifiability Structure

Cross-Level Falsifiability Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.356 Falsifiability via Embedding Spaces

Falsifiability via Embedding Spaces states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried

by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

6.357 Falsifiability in Practice: Multi-Domain Implementation

Falsifiability in Practice: Multi-Domain Implementation states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

7 Source Trace

Exact source bindings, path hashes, quality scorer hooks, and transition records are recorded in the review package manifest. They are kept out of the main prose to avoid turning the document into a ledger dump.